4.6 Similarity and Transformations

Speak **L**ike **A** **M**athematician:

similarity transformation:

similar figures:

Similarity Transformations

A _____ _____ is a dilation, or a composition of rigid motions and a dilation.

Two figures are _____ if and only if a similarity transformation maps one onto the other.

Notation: △ABC ~ △DEF is read "triangle ABC is _____ to triangle DEF."

What Similar Figures Share

Corresponding angles are _____.

Corresponding sides are _____ — their ratios all equal the _____ _____.

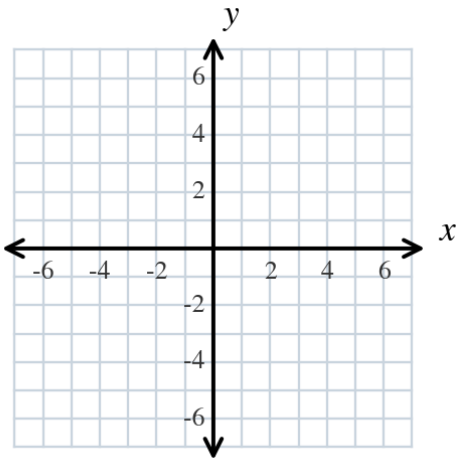
Congruent vs. Similar

Every congruence transformation is a similarity transformation with k = _____.

The reverse is NOT true: similar figures are congruent only when k = _____.

Example 1 – Name the Similarity Transformation

△ABC has A(2, 2), B(4, 6), and C(6, 2). Its image △A′B′C′ has A′(1, 1), B′(2, 3), and C′(3, 1).



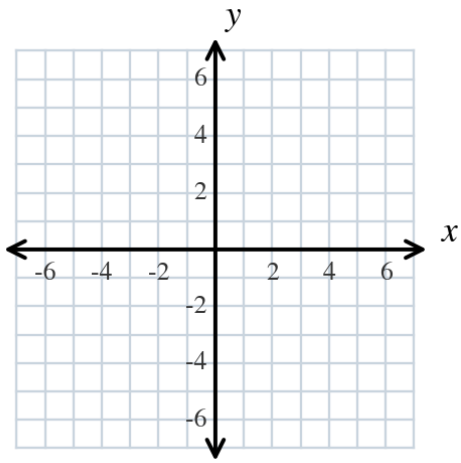
a) Compare matching coordinates. What is the scale factor?

b) Name the similarity transformation.

c) Are the triangles congruent, similar, or both?

Example 2 – Dilation, Then Translation

P(1, 2) is dilated about the origin with k = 2, then translated along ⟨3, −1⟩.



a) Find the coordinates after the dilation.

b) Find the coordinates after the translation.

c) Is the composition a similarity transformation? Explain.

Example 3 – Reflection, Then Dilation

Q(2, −1) is reflected in the y-axis, then dilated about the origin with k = 3.

a) Find the coordinates after the reflection.

b) Find the coordinates after the dilation.

c) Does the order matter here? Try the dilation first.

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Example 4 – Dilation, Then Rotation

R(2, 1) is dilated about the origin with k = 3, then rotated 90° about the origin.

a) Find the coordinates after the dilation.

b) Apply (a, b) ${to} (${M}b, a).

c) Which property is preserved, and which is not?